

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 05

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Organic advantages: often waste products, locally available and cheap; add humus improving structure and water retention; increase soil biota; slow release reduces leaching • Organic disadvantages: nutrient composition cannot be controlled; released slowly so needs long-term planning; bulky with high water content so expensive to transport; cannot be applied to a growing crop; odour / pathogens • Inorganic advantages: composition controlled to match crop needs; nutrients released rapidly; concentrated and easy to apply • Inorganic disadvantages: high energy (fossil fuel) manufacture (Haber process); no organic matter so humus and biota decline; some toxic to worms; high solubility so leached causing eutrophication; raw materials (phosphate rock) finite	6
01.2	• Crops grown in a circulating nutrient solution (inert medium) in greenhouses; nutrient concentrations monitored and adjusted so nutrients are never limiting; abiotic factors controlled • Roots are small (all in contact with nutrients) so more growth goes into the harvested crop; no soil-borne pathogens or weeds; crop stays fresh with roots • Disadvantages: high inputs of nutrients and energy (heating, lighting, pumping); high level of technical knowledge; only economic for high-value crops	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 310-312; AQA spec 3.5.1.1.	

Question	Marking guidance	Marks
02.1	• Warming melts ice shelves that hold back glaciers and lubricates ice flow so land ice reaches the sea faster (Antarctic Peninsula most affected); sea ice loss removes krill habitat and lowers albedo • Raised UV harms planktonic algae, krill and fish / sea urchin larvae in surface waters (land animals protected by fur / feathers) • Disturbance of wildlife, introduction of non-indigenous species and pathogens, fuel and sewage pollution; visitor numbers rising rapidly at a few landing sites • Reserves of coal, oil, gold, uranium, platinum etc; pressure to exploit will rise as reserves elsewhere deplete though currently banned and operations extremely difficult	4
02.2	• Treaty signed by 52 countries including all active there; no military activity (except supporting research), no nuclear explosions or radioactive waste dumping • No commercial mineral exploitation • Fisheries regulated by CCAMLR: krill and some fish exploited, fisheries closed if over-exploited; no commercial exploitation of mammals • Tourism control: no accommodation on land (wastes stay on ships), one ship per site with maximum 100 ashore, trained guides, keep 5 m from wildlife, no litter, cleaned protective clothing to prevent introductions • Waste management: all wastes removed (fuel containers, batteries, plastics, oil); sewage only discharged where sea currents disperse it; old waste removed unless removal causes more damage • Only long-term residents are research scientists and support staff; strict regulation of research impacts	6
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 69-71; AQA spec 3.1.2.3.4.	
03.1	• The nucleus of a stem cell from the endangered species is transferred into an empty egg of a closely related species (southern white rhino) • The embryo is implanted into a surrogate female of the related species which gives birth to a northern white rhino • Very low success rates with wildlife so far; deep-frozen cells could allow clones even after the last individuals die	3
03.2	• Micro-propagation: tissue culture producing many cell clusters from one plant, each grown into a genetically identical plant • Black poplar: cuttings / pollarded stems of pure native trees are rooted to produce many identical trees • Needed because the flowers are pollinated by introduced subspecies producing hybrids, so cloning maintains the pure gene pool (but no genetic diversity)	3
03.3	• Stores seeds of wild plants (especially threatened species and crop wild relatives) so species are not lost if extinct in the wild • Seeds collected worldwide; target of 10 000 seeds per species to maintain a diverse gene pool • Stored dry and refrigerated / frozen in an underground reinforced vault to reduce risk of accident • Seeds germinated periodically to check viability and replenish stocks; can be used for reintroduction / breeding	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 35-37; AQA spec 3.1.2.3.3.	

Question	Marking guidance	Marks
04.1	• Timber: building construction, furniture, tools, telegraph poles, concrete shuttering (oak, beech, mahogany, teak, conifer softwood) • Fuel: fuelwood and charcoal - still the main energy source for most people in LEDCs • Food: fruit, nuts, bushmeat, livestock fodder; crops of forest origin (coffee, cacao, bananas, Brazil nuts); pigs and chickens were forest animals • Fibres: paper (wood pulp cellulose), cotton, viscose / rayon • Medicines: quinine (cinchona), taxol (yew), aspirin (willow); most species not yet researched	5
04.2	• Wood is mainly cellulose which is hard to digest, so carbon stored in wood has a long residence time; total forest carbon (vegetation plus soil) is about double that in the atmosphere • Growing trees sequester CO₂ by photosynthesis and release oxygen; forest soils hold more carbon than the trees (especially boreal) • Planting / protecting forests removes CO₂ and reduces the enhanced greenhouse effect	3
04.3	• Highest transpiration of any ecosystem increases downwind rainfall; interception increases evaporation; deep organic soils retain water and delay runoff so river flow is more even and flooding reduced • Low albedo absorbs sunlight; heat stored and released at night reduces day-night temperature extremes; canopy microclimate: lower light, wind, higher humidity	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 360-364; AQA spec 3.5.3.1.	
05.1	• As sea otters declined (1960-1980) sea urchins increased (to 350) and kelp area fell (to 25) • As otters recovered after 1990 urchins declined and kelp recovered • Otter and kelp trends are positively correlated; urchins negatively correlated with both	3
05.2	• Sea otters are predators of sea urchins so fewer otters allowed the urchin population to rise • Sea urchins graze kelp so the kelp forest was over-grazed and destroyed • Kelp forests are habitat / food for many species including otters (keystone species / trophic cascade); otter recovery restored the balance	3
05.3	<i>Any 2 from:</i> • Over-fishing of sandeels reduced puffins • Over-collection of turtle eggs reduced turtles so jellyfish (their food) increased; over-fishing of sardines / herring (competitors) also increased jellyfish • Loss of pollinating bees (pesticides, wild flower loss) reduces plant reproduction; loss of elephants / large herbivores reduces seed dispersal of trees	2
05.4	• They control abiotic features: beaver dams create ponds colonised by aquatic species; elephants create clearings and keep water holes open • Species that depend on these features (light, water, minerals) cannot survive in their absence	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 23-25; AQA spec 3.1.2.2.	

Question	Marking guidance	Marks
06.1	• Cross-bred offspring are healthier / more productive because the problems caused by homozygous recessive genes in inbred varieties are reduced (heterosis)	1
06.2	• Zebu tolerate heat but have a low milk yield; Ayrshire have a high milk yield but are not heat tolerant • Cross-breeding combined the desirable characteristics: heat tolerance with a high milk yield	2
06.3	<i>Any 2 from:</i> • Replacement of valuable animals that die with genetically identical individuals • Replacement of herds culled during a disease outbreak • Production of large numbers of individuals with desirable characteristics	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-322; AQA spec 3.5.1.2.	
07.1	• Development that meets the needs of the present generation without compromising the ability of future generations to meet their own needs	1
07.2	• Impact depends on total population x per capita impact • With few people, natural processes (forest regrowth, fish recruitment, aerobic decomposition / dilution) recover faster than the resource is used or the waste added • Larger populations exceed the rate of natural recovery: forest and soil do not regenerate, catches exceed MSY, water deoxygenates	3
07.3	• Urban areas are supported by resources brought in from elsewhere; when one area is depleted, exploitation simply moves (timber, fish, minerals, energy) so consumers see no shortage • Rural communities exploit local resources directly so the effects of over-exploitation become obvious quickly, encouraging cultural limits to need	2
07.4	• Climate: warming increases evaporation, cloud and albedo; higher CO₂ increases photosynthesis and sequestration; hydrological cycle: more evaporation gives more precipitation; populations: density-dependent factors return numbers to carrying capacity • In agriculture soil degradation, water shortage and loss of predators would reduce yields (negative feedback limiting the activity) • Humans respond with fertilisers, irrigation and pesticides to maintain yields, further damaging the natural systems that support production	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-372; AQA spec 3.6.1.	
08.1	• MEDCs: market demand and higher prices for the flavour of carnivorous fish (salmon, trout) justify high inputs (fishmeal, cooler climates suit them) • LEDCs: herbivores / omnivores feed on natural vegetation, phytoplankton and wastes so need low inputs; warm climates suit tilapia • Herbivores are more energy-efficient (lower trophic level) so more food per unit area	3
08.2	• Some genetically female (XX) fish are treated with testosterone so they become functional males producing only X-carrying sperm • Crossed with normal females (XX) all offspring are XX females • Female trout meat has a better flavour; in tilapia the reverse (all males) because males grow larger / more efficiently	3

Question	Marking guidance	Marks
08.3	• Salmon need high oxygen: water sprays / weirs aerate; wastes removed to prevent deoxygenation • Constant circular flow keeps fish swimming the same way - fewer collisions / injuries, more muscular fish, higher stocking density • Artificial lighting delays maturation (salmon stop growing at maturity) so larger fish; controls smoltification • Cool sites (Scotland, Norway) because warm water holds less oxygen although warmth increases growth	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 352-355; AQA spec 3.5.2.3.	
09.1	• Mars has low mass so weak gravity and could not retain a dense atmosphere • Very low atmospheric pressure means water would boil / evaporate away even at low temperatures • Greater distance from the Sun gives low insolation and a mean temperature of -60 °C so water is frozen	3
09.2	• Mass great enough for gravity to retain an atmosphere containing the elements needed (C, H, O, N as CO₂, methane, ammonia, water vapour) and to maintain pressure so liquid water does not boil • Distance from the Sun (the 'Goldilocks zone') gives insolation producing temperatures at which water is liquid (0-35 °C over most of the surface) and proteins are not denatured • Too close (Venus) - too hot; too far (Mars) - too cold	3
09.3	• The molten core produces a magnetosphere that deflects the solar wind / charged particles, preventing biologically damaging radiation reaching the surface • The 24-hour rotation limits heating by day and cooling by night, reducing temperature extremes (Venus' slow rotation gives extreme conditions)	2
09.4	• The tilted axis causes seasonal variation in insolation as the Earth orbits the Sun • Producing seasons (temperature, day length) to which species' life cycles (breeding, migration, dormancy) are adapted; the Moon's gravity keeps the tilt stable and creates tides	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 5-7; AQA spec 3.1.1.1.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Succession: pioneers, seres (hydrosere, lithosere, psammosere), modification of abiotic conditions, climax communities controlled by climate; deflected succession and plagioclimaxes • Application: deciding whether to manage (plagioclimax habitats need grazing, mowing, burning, coppicing, pollarding continued) or leave alone (climax habitats); habitat creation planning the stages; controlling scrub / secondary succession; understanding that new habitats change over time • Population dynamics: birth and death rates, density-dependent and independent factors, carrying capacity, sigmoid growth, predator-prey cycles, r- and k-selection, MSY • Application: setting sustainable harvest levels (fisheries, whaling, forestry); recognising k-selected species need strict protection and recover slowly; deciding when culling is needed (deer without wolves, introduced species) or when releases / captive breeding are needed (low breeding rate); predicting effects of reintroducing predators • Monitoring population size, distribution, survival rate and age structure to assess success; adaptation and range of tolerance guiding abiotic management • Evaluation: uncertainties (natural fluctuations, unknown carrying capacity); examples (Yellowstone wolves, Scottish deer, chalk grassland); judgement	9
10.2	• Any taxonomic group of organisms classified together by similarity (eg species, genus, family)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 72-85; AQA spec 3.1.3.	

Question	Marking guidance	Marks
11.1	Discuss the extent to which the resources and services provided by other species justify the conservation of biodiversity. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Resources: wood, fibres, oils, fuels, food; new food species (morama bean, yeheb, perennial maize, Kernza, bison, eland) and their adaptations • Biomimetics: wing-tip feathers, whale tubercles, shark skin, termite mounds, bird bones, gecko adhesion, burrs, lotus, spider silk • Medicines: poppies, cinchona, willow, yew (taxol), Mexican yam, sponges (AZT), sharks; few species yet studied; physiological research (marsupials, hippo, sponges) • Genetic resources: crop wild relatives, Vavilov centres, gene pools and inbreeding • Ecosystem services: atmospheric regulation, hydrological cycle, biogeochemical cycles, soil maintenance; interspecies relationships - food, pollination (Darwin's orchid), seed dispersal, habitat provision • Evaluation: economic vs intrinsic / moral arguments; services unnoticed because of dynamic equilibrium; unknown future value; justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-19; AQA spec 3.1.2.1.	25

Question	Marking guidance	Marks
11.2	Discuss how an understanding of energy flow and the manipulation of food species can be used to increase food production sustainably. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Energy flow: autotrophs and heterotrophs; losses by reflection, transmission, respiration; 100 000 to 1 000 units; few food chains beyond four levels; plant food most productive; grazing animals where crops impossible; pigs on wastes • Energetics: intensive vs extensive, law of diminishing returns, energy subsidies (fertiliser, pesticides, irrigation, machinery, drying, processing, transport), energy ratios, food conversion ratios, dependence on fossil fuels • Control of genetics: asexual reproduction / cuttings / cloning, selective breeding and breeds, cross-breeding and hybrid vigour, AI and embryo transfer • GM: Roundup Ready soya, Bt crops, golden rice, omega-3 oilseed rape; advantages; risks (gene transfer, antibiotic markers, allergies, costs to LEDC farmers, loss of local diversity) • Sustainability links: energy dependence, trophic level of diet, feeding 12 billion people • Justified conclusion on the balance between productivity, efficiency and sustainability Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-328; AQA spec 3.5.1.2.	25